

Building Technology

Engineering Mathematics I for Building Technology

Assignment 1

Instructions: Answer all questions. Show formula, substitution, calculation, units and technical interpretation.

Section A: Formula and Concept Questions

1. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$y=mx+c$$

2. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\tan\theta=\frac{\mathrm{rise}}{\mathrm{run}}$$

3. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$V=\frac{1}{3}Ah$$

4. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$C=\pi d$$

5. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\frac{dy}{dx}=f'(x)$$

Section B: Applied Problems

6. Solve a building technology calculation problem involving roof pitch. Show all working and write one technical interpretation. (10 marks)
7. Solve a building technology calculation problem involving stair gradients. Show all working and write one technical interpretation. (10 marks)
8. Solve a building technology calculation problem involving excavation volumes. Show all working and write one technical interpretation. (10 marks)
9. Solve a building technology calculation problem involving cost curves. Show all working and write one technical interpretation. (10 marks)
10. Solve a building technology calculation problem involving construction graphs. Show all working and write one technical interpretation. (10 marks)
11. Solve a building technology calculation problem involving roof pitch. Show all working and write one technical interpretation. (10 marks)
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19. Solve a building technology calculation problem involving cost curves. Show all working and write one technical interpretation. (10 marks)
20. Solve a building technology calculation problem involving construction graphs. Show all working and write one technical interpretation. (10 marks)

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.